

Versal VCK190: 80-Pipeline Packet-Switched Architecture

PROCESSING SYSTEM (PS)

ARM Cortex-A72 Quad-Core @ 1.5 GHz

XRT Host Application

RAMBO
Madgraph PSP
generation

Benchmark
metrics

DDR4 Memory (32 GB)

- Input: 8 PSP buffers (80 bytes/PSP)
- Output: 8 result buffers (4 bytes/result)

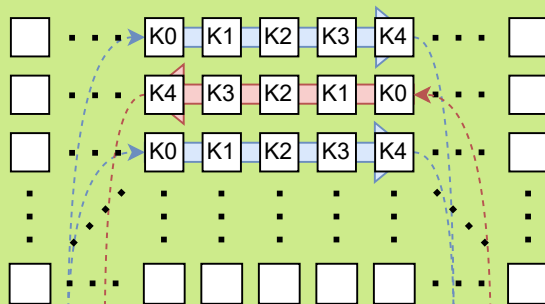
Bandwidth: 32 GB/s

NOC

AXI-MM (DDR)
DDR Read/Write
32 GB/s

AI ENGINE ARRAY (AIE)

400 Compute Tiles: 50 columns × 8 rows @ 1.25 GHz



I O

Each Pipeline (5 tiles):
K0(pkt) → K1 → K2 → K3 → K4(pkt)
Latency: 80 μ s | Throughput: 12.5k PSP/s

O I

AXIS
PLIO 500 MHz
32b (Packet)

PROGRAMMABLE LOGIC (PL)

Packet Generation & Parsing Layer @ 500 MHz

MM2S_Pkt [0..8]

- Burst-read PSPs
- Generate headers
- Stream to PLIO

S2MM_Pkt [0..8]

- Receive from PLIO
- Extract ME² results
- Burst-write DDR